Literature Mining for Rare Disease Phenotype Genotype Associations

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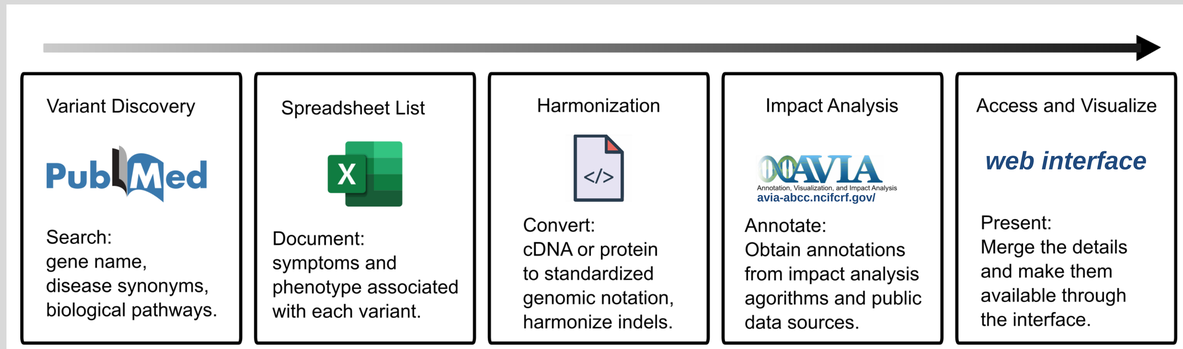
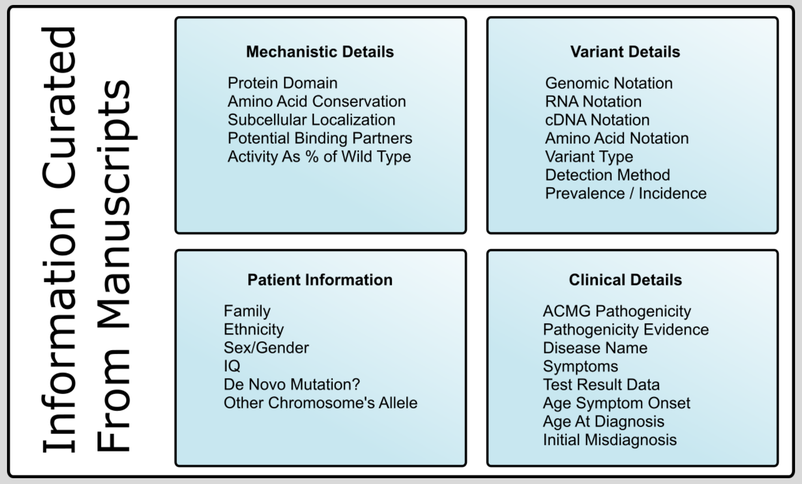
Abstract

Although individually uncommon, collectively, rare diseases affect 6-8% of the world's population[1]. Diagnosing rare diseases through phenotypes is a difficult task because symptoms overlap based solely on phenotype - there may be multiple causes for a single disease and a single cause may be associated with multiple diseases. Genomic investigations are leading to precise molecular level characterization allowing for a systematic discovery of therapies that either target a specific disease or, more broadly, multiple related diseases.
Currently, text mining techniques have lower accuracy and more coverage gaps than manual curation, but the potential for higher productivity. Recently, there has been some progress in developing text mining applications to tackle this enormous problem. However, there is still a need to assess these text mining applications and integrate the findings with data. Towards these ends, we manually created a rare disease variant data set which can be used to test and refine a text mining algorithm. We developed a manual curation workflow which incorporated searching for genetic variants on PubMed, listing symptoms and phenotypes associated with each variant, converting cDNA or protein to standardized genetic notation, obtaining annotations from public data sources, and presenting the details in an online interface for future release. Our project aims include summation of pathogenic variant frequency in populations to estimate birth prevalence for each rare disease. We created test data sets of known accuracy, coverage, and genotype-phenotype associations that can be used to validate text mining approaches. Enhanced text mining will significantly decrease the amount of time necessary to gather data to molecularly characterize a rare disease and render it possible to mine rare disease phenotype-genotype associations for the more than 7,000 rare disease genes in a timely fashion.

Methods

Manual curation, harmonization, and presentation methods (Figure 1) are as follows. The curator searched PubMed for the gene name, disease synonyms, and biological pathways of relevance. The curator created a spreadsheet list of all SLC6A8 (associated with Creatine Transport Deficiency) variants, documenting the symptoms and phenotype associated with each variant. The types of information recorded from the manuscripts are shown in Figure 2. The information collected was assessed for accuracy by comparing to the ClinVar database of genetic variants. 100% of the ClinVar variants noted as having been published had been found by the manuscript curation process. During the harmonization step, the cDNA or protein format of the variant was converted to standardized genomic notation using a custom developed software. The results were then manually checked and harmonized. Details were used to improve validation accuracy and to improve the software's sensitivity and specificity. In cases where the published variant name did not match the standard notation, the manuscript's text was investigated for context. For example when a variant's amino acid was referred to as both L and K, context of participation in a leucine zipper led the curator to conclude L. Error prone abbreviations include N, Q, D, E, and L. AVIA was employed to obtain annotations from public data sources and analyze the impact. Overlap between the manually curated variants and those in the ClinVar database was assessed. Lastly, the results are to be included on a future-released website that will be available to the public to access and visualize the rare disease variants on 2- and 3-D protein structures (Figure 3).





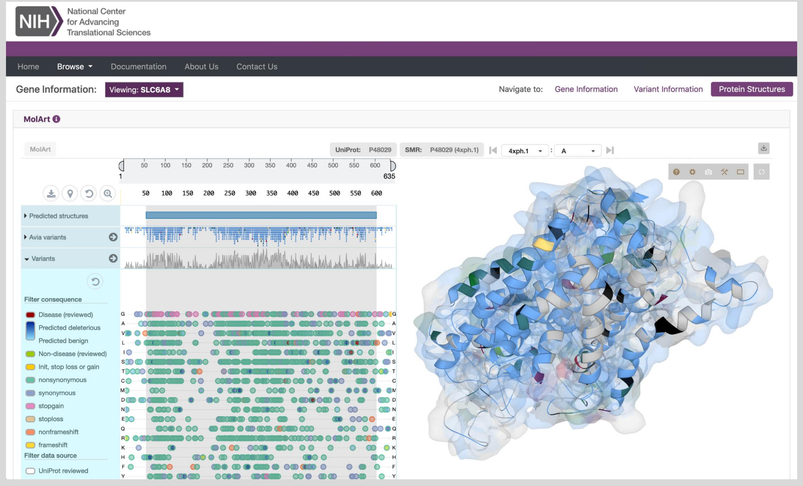


Figure 1: Tools workflow for curation of variants from PubMed, harmonization, and creation of a publicly available 3-D variant display.

Figure 2: Information curated from PubMed manuscripts for researchers and clinicians in table format.

Figure 3: The website displays the 3-D protein structure and a sortable list of known genetic variants. Clicking on the variant highlights it on the 3-D structure.

Results

Overall, 190 SCL6A8 variants were curated from PubMed. Of these, 91 were pathogenic. Single nucleotide missenses were most likely to be properly converted, while insertions, deletions, frameshift, and intronic variants are more susceptible to improper conversion during harmonization.

The ClinVar database had 278 variants with 50 or fewer base pairs. ClinVar had 51, or 27%, of the total curated variants. ClinVar had 16% of the pathogenic variants curated from PubMed (see Figure 4 below).

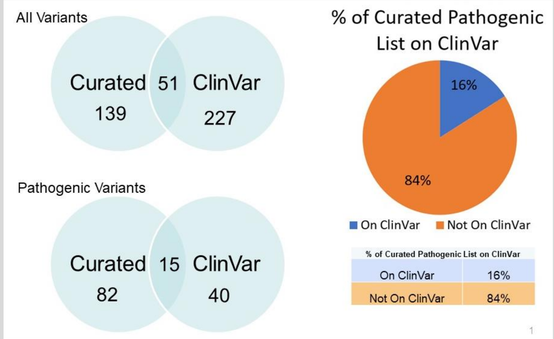


Figure 4: Overlap between the manually curated variants from PubMed and the ClinVar database as of March 2020.

Future Directions:

The curated SLC6A8 data set can serve as a test data set of known coverage and accuracy to train text mining algorithms to mine PubMed for rare disease gene variants. The curation process including initial review, standardization and validation using context can improve sensitivity and specificity of such automated algorithms and help expand the extraction of gene-variant-pathogenicity relationships for the more than 7000 known rare diseases. We are actively exploring collaborations with text mining experts and have open positions to support the challenging task.

References:
1 Dawkins, H. J. S. *et al.* Progress in Rare Diseases Research 2010-2016: An IRDiRC Perspective. *Clin Transl Sci* **11**, 11-20, doi:10.1111/cts.12501 (2018).

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